

ALESSANDRO FEDERICI

AUTOMATION / AI DEVELOPER

CONTACT

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Bologna, Italy

Website (Portfolio) and Website

LinkedIn

EDUCATION

2016 August
ITIS ODONE BELLUZZI

- High School in Electronics

SKILLS

AI & Automation
- n8n (self-hosted)
- Make (Integromat)
- Zapier
- Workflow Automation Design
- AI Agents (Voice & Chat)

AI / LLM
- OpenAI API (GPT, Embeddings)
- Retrieval-Augmented Generation (RAG)
- Prompt Engineering
- LLM-based Automation Systems
- LangChain (basic usage)
- LangGraph (concepts)

Backend & Data
- REST APIs & Webhooks
- Supabase (PostgreSQL, pgvector)
- Airtable

PROFILE

AI Automation Developer specialized in building scalable automation systems and AI-driven applications using tools like n8n, Make, and modern LLM APIs.
I design and develop end-to-end solutions that integrate APIs, CRMs, databases, and AI models to automate business processes and reduce manual workload. My work includes AI agents (voice and chat), multi-step workflows, and retrieval-based systems (RAG) for structured knowledge access.

I also leverage AI-assisted development tools (vibe coding) such as Claude Code, OpenAI Codex, and Google Antigravity to rapidly build custom applications, MVPs, and web platforms.

Additionally, I design and deploy modern websites and landing pages using tools like Framer or custom AI-assisted development workflows, focusing on performance, scalability, and clean user experience.
I have hands-on experience in self-hosting and scaling automation infrastructures (n8n) using VPS environments, Docker, and deployment platforms like Coolify.

My approach is strongly execution-focused: I build practical, reliable systems that solve real operational problems, not just prototypes.

I continuously stay up to date with emerging technologies and evolving software stacks, with a strong focus on identifying the most efficient tools and approaches, especially within the fast-moving AI ecosystem.

WORK EXPERIENCE

Freelance, Remote 2021 - PRESENT

Automation / AI Developer

- Designed and implemented end-to-end automation systems using n8n (self-hosted), Make (Integromat), and API integrations
- Built AI-powered workflows using OpenAI (GPT, embeddings) for content generation, data processing, and decision-making logic
- Developed multi-step automation pipelines integrating webhooks, REST APIs, and external services
- Implemented RAG-based architectures using vector databases (Supabase / pgvector) for structured knowledge retrieval and AI-assisted querying
- Built AI agents (voice and chat) for real-world applications, including appointment booking systems with dynamic availability handling and external tool integrations
- Leveraged AI-assisted development tools (Claude Code, OpenAI Codex, Google Antigravity) to rapidly develop custom applications, internal tools, and MVPs
- Designed and developed modern websites and landing pages using Framer and AI-assisted coding workflows, optimizing for performance and user experience
- Managed self-hosted automation infrastructures using VPS environments, Docker, and Coolify, ensuring scalability and reliability of workflows

- CRM Integrations (Monday, ClickUp, GoHighLevel)
- Data Scraping (HTTP, Apify)
- Data Processing Pipelines

Development
- Python
- JavaScript (basic)
- API Integrations
- MVP & Rapid Prototyping (Vibe Coding)

Infrastructure & DevOps
- VPS Management
- Docker
- Coolify
- Self-hosting Automation Systems

Web & Frontend
- Framer
- WordPress
- AI-assisted Web Development

LANGUAGES

- English (Advanced)
- Italian (Native)

- Built document and data extraction systems (e.g. invoices, reports) using AI models and structured storage solutions (Airtable, Google Drive)
- Integrated multiple third-party platforms and services to enable seamless data flow across business tools and ecosystems
- Optimized and maintained automation systems to reduce manual operations, improve performance, and increase process reliability
- Developed automated content pipelines including web scraping, AI rewriting, media handling, and publishing to CMS platforms (e.g. WordPress) and social media
- Integrated and automated CRM systems (Monday, ClickUp, GoHighLevel) to streamline lead management, task automation, and customer workflows

Sele Lift 2016 - 2017

Laboratory Technician

- Company specializing in the production of mobile elevators and freight lifts. With more than 45,000 elevators installed worldwide, Sele is currently present on 5 continents and in 35 countries.

COURSES / CERTIFICATIONS

- Building Generative AI-Powered Applications with Python
- Python Project for AI and Application Development
- Python for Data Science and AI, IBM (Coursera)
- Generative AI: Prompt Engineering, IBM (Coursera)
- Generative AI Essentials, IBM (Coursera)
- Artificial Intelligence Essentials V2, IBM (Coursera)
- Software Engineering Essentials, IBM (Coursera)
- Introduction to HTML, CSS, & JavaScript, IBM (Coursera)
- Make Advanced

For major details click the course link.
To view up-to-date certifications here's my Credly page

PROJECTS (A FEW)

- **AI RAG System (2-Step Retrieval)** ---> **Full Details Here** and **Here**

Built a retrieval-augmented generation system using Supabase (pgvector) and OpenAI embeddings. Implemented a two-step retrieval pipeline (document-level + chunk-level) with reranking for improved accuracy and contextual responses.

- **AI Voice Booking Agent** ---> **Full Details Here**

Developed a voice AI agent using ElevenLabs integrated with n8n to handle appointment bookings. The system dynamically retrieves availability, manages user data, and confirms bookings via external tools.

- **AI Content Automation Pipeline** ---> **Full Details Here** and **Here**

Created an automated workflow that scrapes articles, rewrites content using AI, and publishes it to WordPress. Extended the system to auto-generate social media posts with captions and media assets.

- **AI Document Processing System** ---> **Full Details Here**

Built an automation to extract structured data from documents (e.g. invoices) using AI models, storing results in Airtable and integrating with Google Drive and Slack for notifications.

- **AF Outreach (Lead Generation WebApp) ---> Full Details Here**

Built an automated lead generation system that collects and enriches business data, processes it through AI, and prepares structured outreach workflows for sales and marketing use.

- **Business Analyzer (AI + Scraping + PDF Reports) ---> Full Details Here**

Developed an AI-powered system that analyzes company websites by scraping multiple pages, generating structured insights (including SWOT analysis), and producing downloadable PDF reports. Includes caching system to avoid redundant processing.

- **AF Speech (Speech to text App) → Full Details Here**

AF Speech is a Windows desktop application designed to eliminate manual typing completely, allowing users to write anywhere simply by speaking.

ADDITIONAL EXPERIENCE

- **Jenia n8n Hackathon - Sassuolo, Italy**

Built a functional automation prototype using n8n, integrating APIs and real-time data flows
Focused on rapid development, workflow orchestration, and problem-solving under time constraints